

Sumaiya Sultana Shaik

GitHub — LinkedIn

sultanasumaiya623@gmail.com — +91 97015 28467
Hyderabad, India

Computer Science Engineer with a strong data structures and algorithms foundation (300+ problems solved) and 6 months of production software engineering experience at Techolution, building backend services, REST APIs, and access-control systems for a live multi-tenant SaaS platform. Comfortable owning work across the stack, including API design, database modeling, cloud deployment (GCP, Docker), and React/TypeScript frontends, with hands-on experience integrating LLMs, RAG, and agentic workflows (LangChain, LangGraph) into production systems. Published researcher in applied NLP and ML, looking for Software Engineering, Backend, Full Stack, or Applied AI Engineer roles where I can own systems end to end.

EDUCATION

- Keshav Memorial Institute of Technology** Hyderabad, India
B.Tech in Computer Science Engineering — CGPA: 7.10/10 *Nov 2022 to June 2026*

SKILLS

- **Languages:** Python, Java, TypeScript, SQL
- **Backend & System Design:** REST APIs, GraphQL, FastAPI, RBAC, OAuth 2.0, microservices, multi-tenant architecture
- **AI/LLM:** LangChain, LangGraph, RAG systems, agentic workflows, prompt engineering, scikit-learn
- **Databases:** PostgreSQL, MongoDB
- **Cloud & DevOps:** GCP (Cloud Run, Cloud Storage), Docker, Git/GitHub, CI/CD
- **Frontend:** React, TypeScript, Tailwind CSS
- **Testing & Process:** pytest, Postman, Agile/Scrum
- **Concepts:** Data structures and algorithms (300+ problems solved on LeetCode/GeeksforGeeks), object-oriented design

EXPERIENCE

- Techolution** Hyderabad, India
Software Engineering Intern (Python, FastAPI, React, TypeScript, Docker, GCP) *Oct 2025 to Apr 2026*
 - Designed and implemented RBAC for a multi-tenant SaaS platform, including org-level data isolation, per-resource permission checks, and in-org sharing, enforced at both the FastAPI service layer and the React/TypeScript UI; independently designed the KYC onboarding flow for organizations and applications.
 - Built the OAuth 2.0 connector service and a 3-tier automated reminder system (48h/24h/6h) powering a sales-prep tool and an email triage product, removing a manual follow-up step from the sales team's workflow; wrote typed Python SDK wrappers around external REST APIs (Google Drive, SharePoint, Linear, Calendar) with retry logic and structured logging.
 - Designed prompt architectures and orchestration logic for a multi-agent document-generation system (Request Analysis, HVPD, and Statement of Work workflows across two contract templates) with human-in-the-loop review, cutting proposal turnaround from hours to minutes; led client workshops to translate business requirements into buildable use cases.
 - Investigated and resolved production incidents through root-cause analysis and validation hardening, reducing recurring API failure rate by ~45%; delivered real-time LLM response streaming via Server-Sent Events, cutting perceived latency by ~40%; wrote pytest suites covering validation logic and API contracts, and collaborated daily with product, QA, and pre-sales stakeholders in 2-week Agile sprints.

PROJECTS

- **Privacy Risk Intelligence Platform** [GitHub] 2026
 - Built a 3-layer privacy-risk analysis system (FastAPI + React): a spaCy/Microsoft Presidio NLP layer detecting 8+ PII entity types, a Random Forest classifier scoring text as LOW/MEDIUM/HIGH risk (~89% accuracy on a held-out test set), and a LangChain-orchestrated LLM layer that generates contextual remediation advice and privacy-safe rewrites, invoking the LLM selectively (high-risk or complex cases) rather than on every request. Optimized for sub-2-second end-to-end response time and covered by unit/integration tests. Methodology published as a peer-reviewed paper (see Publications).
 - *Stack: Python, FastAPI, scikit-learn, spaCy, Presidio, LangChain, MongoDB, React, Tailwind CSS.*
- **WealthWise: AI Financial Planning Platform** [GitHub] [Demo] 2025
 - Built a full-stack personal finance assistant using an agent-based architecture and a Retrieval-Augmented Generation (RAG) layer over transaction and market data to ground LLM-generated advice in real user data, covering goal-based financial planning, investment recommendations, expense categorization, and portfolio performance review; built the REST API backend and React frontend end-to-end, and designed a synthetic-data pipeline to work around limited access to real financial data.
 - *Stack: Python, FastAPI, LangChain/agents, RAG, PostgreSQL, React.*

PUBLICATION

- **A Hybrid AI Framework for Privacy Risk Analysis in Digital Footprints Using NLP, ML, and LLMs:**
Peer-reviewed paper on automated PII detection, ML-based risk classification, and LLM-driven remediation using a hybrid multi-layer architecture. *IRJET, Vol. 13, Issue 5, May 2026.* [Journal Link]